Refinement via Regeneration: Enlarging Modification Space Boosts Image Refinement in Unified Multimodal Models

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Research Questions

- How can unified multimodal models perform image refinement without the constraints of pixel-level preservation and intermediate editing instructions?
- Can reformulating image refinement as conditional regeneration rather than editing improve prompt-image alignment?
- What training paradigms enable better semantic correction in text-to-image generation through enlarged modification spaces?

Core Idea

Current unified multimodal models refine images through editing instructions, but these instructions are often incomplete and pixel-level preservation unnecessarily restricts what can be changed. The key insight is to treat refinement as conditional regeneration instead of editing - generate a new image conditioned on the target prompt and semantic tokens of the initial image, without forcing pixel-level consistency. This enlarges the modification space, allowing the model to make broader structural changes for better prompt alignment rather than being constrained to local edits.

Key Formula

RvR Flow Matching Loss

$$L_{FM} = E\left[\left\|v_\theta(x_t, T_{prompt}, Z_{ViT}) - (x_1 - x_0)\right\|_2^2\right]$$

x_t are noisy VAE tokens, T_prompt is target prompt, Z_ViT are semantic visual tokens from initial image, v_θ is velocity field prediction. Unlike RvE, this excludes Z_VAE (pixel-level tokens) from conditioning, enabling semantic-level rather than pixel-level alignment.

Key Figure

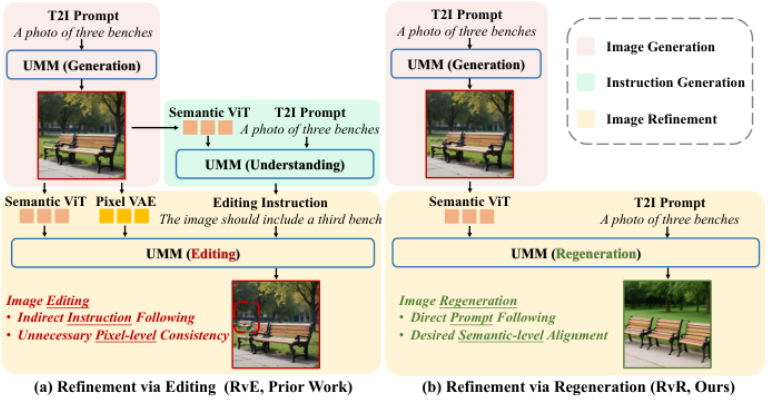

Fig. 3: Comparison between (a) prior Refinement via Editing (RvE) and (b) our Refinement via Regeneration (RvR). RvE requires precise instruction generation and content consistency for unedited regions, while RvR discards these unnecessary constraints, enlarging modification space for better refinement.

producing three natural, well-composed benches—even if achieving this requires broader structural changes beyond localized edits.

Motivated by these observations, we propose *Refinement via Regeneration* (RvR), a novel framework that reformulates image refinement from a regeneration perspective, as illustrated in Fig. 3(b). Rather than relying on intermediate editing instructions or enforcing pixel-level consistency with the input image, RvR conditions directly on the target prompt together with the semantic tokens of the initial image, treating refinement as another round of image generation. Removing editing-specific constraints substantially enlarges the effective modification space, allowing the model to revise any regions that hinder prompt satisfaction. As shown in Fig. 3(b), RvR produces refined images that are semantically correct, spatially coherent, and better aligned with the target prompt.

This regeneration-based refinement mechanism is supported by a training scheme that enlarges the effective modification space from both data and pipeline perspectives. At the data level, we construct supervision from independently generated T2I samples with varying degrees of prompt alignment, rather than from editing pairs that enforce strict content consistency. Consequently, the aligned image need not be an edited version of the misaligned image; the supervision emphasizes semantic correction toward the target prompt without imposing pixel-level correspondence. At the pipeline level, RvR further simplifies the refinement pipeline by discarding pixel-level VAE conditioning and relying only on semantic ViT representations of the input image. This design allows the model to revise image content guided by high-level semantics, instead of being biased toward appearance preservation.

Extensive experiments across multiple benchmarks demonstrate that RvR consistently delivers substantial gains in T2I generation, with particularly strong

Figure 3: Clearly contrasts RvE vs RvR paradigms, showing how RvR eliminates editing instructions and pixel-level constraints for enlarged modification space.

Limitations

- Requires careful balance between semantic preservation and modification freedom during training
- Performance depends on quality of semantic visual encoder (ViT) representations
- May occasionally discard semantically compatible elements from initial image that should be preserved
- Training data construction relies on independently generated samples which may not cover all refinement scenarios